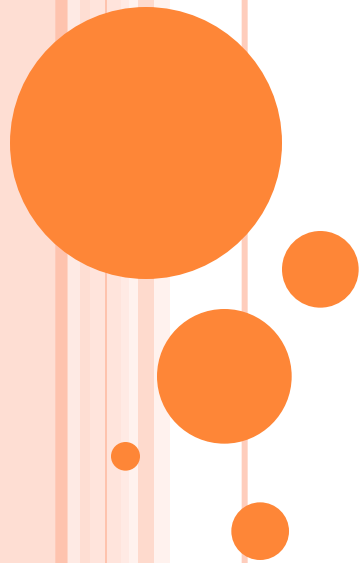


BOOLEAN ALGEBRAS AND BOOLEAN FUNCTIONS



BOOLEAN ALGEBRAS AND BOOLEAN FUNCTIONS

- **George Boole introduced Boolean algebras in 1854.**
- **In 1938 Claude Shannon proved that a two-valued binary Boolean algebra can describe the operations of two-valued electrical switching circuits.**
- **In modern times Boolean algebras and Boolean functions are indispensable in the design of computer chips and digital circuits.**



BOOLEAN ALGEBRA

Definition 1. A **Boolean algebra** is a structure $(A, \wedge, \vee, \bar{}, 0, 1)$ described by the axioms:

- $|A| \geq 2$, A contains at least two elements:
0 and 1, $0 \neq 1$
- $\wedge, \vee$ - binary operations
- $\bar{}$ - unary operation
- 0 - zero element:
 $x \wedge 0 = 0 \wedge x = 0$ and
 $x \vee 0 = 0 \vee x = x, \forall x \in A$
- 1 - unity element:
 $x \wedge 1 = 1 \wedge x = x$ and $x \vee 1 = 1 \vee x = 1,$
 $\forall x \in A$
- 0- the first element, 1- the last element
 $x \wedge \bar{x} = 0$ and $x \vee \bar{x} = 1, \forall x \in A$
duble negation: $\bar{\bar{x}} = x, \forall x \in A$
- duble negation: $\bar{\bar{x}} = x, \forall x \in A$
- commutativity:
 $x \wedge y = y \wedge x$ and $x \vee y = y \vee x$
- associativity:
 $x \wedge (y \wedge z) = (x \wedge y) \wedge z (= x \wedge y \wedge z)$ and
 $x \vee (y \vee z) = (x \vee y) \vee z (= x \vee y \vee z)$
- distributivity:
 $x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z)$ and
 $x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z),$
- idempotency:
 $x \wedge x = x$ and $x \vee x = x, \forall x \in A$
- De Morgan's laws:
 $\overline{x \wedge y} = \bar{x} \vee \bar{y}$ and $\overline{x \vee y} = \bar{x} \wedge \bar{y},$
- absorption laws:
 $x \wedge (x \vee y) = x$ and $x \vee (x \wedge y) = x$

EXAMPLE 1

The binary Boolean algebra is $B = (B_2 = \{0, 1\}, \wedge, \vee, \bar{}, 0, 1)$, where:

$\vee$	0	1
0	0	1
1	1	1

$\wedge$	0	1
0	0	0
1	0	1

x	$\bar{x}$
0	1
1	0

Alternative symbols used in *binary Boolean algebra* are:

$*$ (for $\wedge$) and $+$ (for $\vee$).



Example 2.

The structure $(F_P, \wedge, \vee, \neg, F, T)$ is a Boolean algebra:

- $|F_P| \geq 2$, F_P is the set of all well-formed propositional formulas;
- $F, T \in F_P$, where F (false), T (true) are the truth values;
- $\wedge$ (conjunction), $\vee$ (disjunction);
- $\neg$ (negation);

Example 3.

The structure $(P(X), \cap, \cup, C_X, \emptyset, X)$ is a Boolean algebra.

- $P(X)$ is the set of all subsets of X .
- $C_X(A) = X - A$, denoted $C(A)$ is the complementary set of A with respect to X



BOOLEAN FUNCTIONS

Let $B = (B_2, \wedge, \vee, \bar{}, 0, 1)$ be the binary Boolean algebra, $B_2 = \{0, 1\}$ și $n \in \mathbb{N}^*$.

A Boolean function of n variables is a function $f : (B_2)^n \rightarrow B_2$

defined as follows:

1. The projection function: $P_i : B_2^n \rightarrow B_2$, $P_i(x_1, \dots, x_i, \dots, x_n) = x_i$,
is a Boolean function.
2. If $f, g : B_2^n \rightarrow B_2$ are Boolean functions then $f \wedge g$, $f \vee g$, $\bar{f}$
are Boolean functions, where:
$$(f \wedge g)(x_1, \dots, x_n) = f(x_1, \dots, x_n) \wedge g(x_1, \dots, x_n);$$
$$(f \vee g)(x_1, \dots, x_n) = f(x_1, \dots, x_n) \vee g(x_1, \dots, x_n);$$
$$\bar{f}(x_1, \dots, x_n) = \overline{f(x_1, \dots, x_n)}.$$
3. Any Boolean function can be obtained by a finite application of the rules 1 and 2.

THEOREMS

Theorem 1.

$\forall n \in \mathbb{N}^*$, there exist 2^{2^n} Boolean functions of n variables.

Theorem 2.

The structure $(\text{FB}(n), \wedge, \vee, \bar{}, f_0, f_{2^{2^n}-1})$ is a Boolean algebra, |

where $\text{FB}(n)$ is the set of all Boolean functions of $n \in \mathbb{N}^*$ variables,

the operations $\wedge, \vee$ and $\bar{}$ are defined in the above definition and

$f_0(x_1, \dots, x_n) = 0$, $f_{2^{2^n}-1}(x_1, \dots, x_n) = 1$ are the constant functions: 0 and 1.



EXAMPLE 4

For $n=1$, there are $4 = 2^2 = 2^{2^1} = 2^{2^n}$ Boolean functions of one variable:

x	$f_0(x)=0$	$f_1(x) = x$	$f_2(x)= \bar{x}$	$f_3(x)=1$
0	0	0	1	1
1	0	1	0	1



Example 5.

There are $2^{2^2} = 2^4 = 16$ Boolean functions of 2 variables

x	y	$f_0(x,y)$	$f_1(x,y)$	$f_2(x,y)$	$f_3(x,y)$	$f_4(x,y)$	$f_5(x,y)$	$f_6(x,y)$	$f_7(x,y)$
		<i>contradiction</i>	$x \wedge y$	$x \wedge \bar{y}$	x	$\bar{x} \wedge y$	y	$x \oplus y$	$x \vee y$
0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	1	1	1	1
1	0	0	0	1	1	0	0	1	1
1	1	0	1	0	1	0	1	0	1

x	y	$f_8(x,y)$	$f_9(x,y)$	$f_{10}(x,y)$	$f_{11}(x,y)$	$f_{12}(x,y)$	$f_{13}(x,y)$	$f_{14}(x,y)$	$f_{15}(x,y)$
		$x \downarrow y = \bar{x} \wedge \bar{y}$	$x \leftrightarrow y$ $x \oplus y$	$\bar{y}$	$y \rightarrow x = \bar{y} \vee x$	$\bar{x}$	$x \rightarrow y = \bar{x} \vee y$	$x \uparrow y = \bar{x} \vee \bar{y}$	<i>tautology</i>
0	0	1	1	1	1	1	1	1	1
0	1	0	0	0	0	1	1	1	1
1	0	0	0	1	1	0	0	1	1
1	1	0	1	0	1	0	1	0	1

EXAMPLE 5 (CONTD.)

Each function has an inverse function obtained as its negation.

Pairs of inverse functions:

$$(f_0, f_{15}), (f_1, f_{14}), (f_2, f_{13}), (f_3, f_{12}), (f_4, f_{11}), (f_5, f_{10}), (f_6, f_9), (f_7, f_8).$$

We use the following symbols for operations:

$\downarrow$ (nor) – Pierce's function, $x \downarrow y = \overline{(x \vee y)}$

$\uparrow$ (nand) – Schaffer's function, $x \uparrow y = \overline{(x \wedge y)}$

$\rightarrow$ (logical implication), $x \rightarrow y = \bar{x} \vee y$

$\oplus$ („xor”), $x \oplus y = \overline{(x \leftrightarrow y)} = (\bar{x} \wedge y) \vee (x \wedge \bar{y})$

$\otimes$ („nxor”), $\leftrightarrow$ (logical equivalence),

$x \otimes y = x \leftrightarrow y = (x \rightarrow y) \wedge (y \rightarrow x) = (x \wedge y) \vee (\bar{x} \wedge \bar{y})$



NOTATIONS

For a uniform representation of variables and their negations we use the following notation:

$$x^{\alpha} = \begin{cases} x, & \text{if } \alpha = 1 \\ \bar{x}, & \text{if } \alpha = 0 \end{cases}, \quad x \in \{0, 1\},$$

$$x^{\alpha} = \begin{cases} 1, & \text{if } x = \alpha \\ 0, & \text{if } x \neq \alpha \end{cases}, \quad x, \alpha \in \{0, 1\}$$

For $x, \alpha \in \{0, 1\}$:

$$x^0 = \bar{x}, \quad x^1 = x \quad \text{and} \quad 0^0 = \bar{0} = 1; \quad 0^1 = 0; \quad 1^0 = \bar{1} = 0; \quad 1^1 = 1$$



CANONICAL FORMS OF BOOLEAN FUNCTIONS

Theorem 3.

A Boolean function $f : (B_2)^n \rightarrow B_2$, $n \in \mathbb{N}^$ can be transformed into two equivalent forms:*

1. *disjunctive canonical form (DCF):*

$$(1) \quad f(x_1, \dots, x_n) = \bigvee_{(\alpha_1, \dots, \alpha_n) \in B_2^n} (f(\alpha_1, \dots, \alpha_n) \wedge x_1^{\alpha_1} \wedge \dots \wedge x_n^{\alpha_n})$$

2. *conjunctive canonical form (CCF):*

$$(2) \quad f(x_1, \dots, x_n) = \bigwedge_{(\alpha_1, \dots, \alpha_n) \in B_2^n} (f(\alpha_1, \dots, \alpha_n) \vee x_1^{\overline{\alpha_1}} \vee \dots \vee x_n^{\overline{\alpha_n}})$$



CANONICAL FORMS OF BOOLEAN FUNCTIONS



M. Lupea

Theorem 4.

A Boolean function $f: (B_2)^n \rightarrow B_2$ is unique determined by its values

$f(\alpha_1, \dots, \alpha_n)$, where $(\alpha_1, \dots, \alpha_n) \in B_2^n$:

1. *disjunctive canonical form (DCF):*

$$(1) \Leftrightarrow (1') \quad f(x_1, \dots, x_n) = \bigvee_{(\alpha_1, \dots, \alpha_n) \in B_2^n \text{ and } f(\alpha_1, \dots, \alpha_n)=1} (x_1^{\alpha_1} \wedge \dots \wedge x_n^{\alpha_n})$$

2. *conjunctive canonical form (CCF):*

$$(2) \Leftrightarrow (2') \quad f(x_1, \dots, x_n) = \bigwedge_{(\alpha_1, \dots, \alpha_n) \in B_2^n \text{ and } f(\alpha_1, \dots, \alpha_n)=0} (x_1^{\overline{\alpha_1}} \vee \dots \vee x_n^{\overline{\alpha_n}})$$

Example 6.

For $n=2$, the disjunctive canonical forms of the functions f_8, f_{13}, f_6, f_{12} from Example 5

$$f_8(x, y) = (1 \wedge x^0 \wedge y^0) \vee (0 \wedge x^0 \wedge y^1) \vee (0 \wedge x^1 \wedge y^0) \vee (0 \wedge x^1 \wedge y^1) = \bar{x} \wedge \bar{y}$$

$$f_{13}(x, y) = (1 \wedge x^0 \wedge y^0) \vee (1 \wedge x^0 \wedge y^1) \vee (0 \wedge x^1 \wedge y^0) \vee (1 \wedge x^1 \wedge y^1) = (\bar{x} \wedge \bar{y}) \vee (\bar{x} \wedge y) \vee (x \wedge y)$$

$$f_6(x, y) = (0 \wedge x^0 \wedge y^0) \vee (1 \wedge x^0 \wedge y^1) \vee (1 \wedge x^1 \wedge y^0) \vee (0 \wedge x^1 \wedge y^1) = (\bar{x} \wedge y) \vee (x \wedge \bar{y})$$

$$f_{12}(x, y) = (1 \wedge x^0 \wedge y^0) \vee (1 \wedge x^0 \wedge y^1) \vee (0 \wedge x^1 \wedge y^0) \vee (0 \wedge x^1 \wedge y^1) = (\bar{x} \wedge \bar{y}) \vee (\bar{x} \wedge y)$$

Note that the functions' indices, represented in binary using 4 binary digits, are the binary numbers obtained reading the corresponding column.

For $f_{12}(x, y)$, $12=1100_{(2)}$ and the column of the function's values is: $\begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}$.



Remarks:

- 1.The disjunctive canonical form is recommended when the function has more values 0 than 1.
- 2.The conjunctive canonical form is recommended when the function has more values 1 than 0.

Example 8. Write the corresponding canonical forms of f_2, f_5, f_3, f_0 .

Formula (1') will be used for f_2, f_5 and formula (2') will be used for $f_3(x, y), f_0(x, y)$:

$$f_2(x, y) = (x^1 \wedge y^0) = x \wedge \bar{y} \quad \text{(DCF)}$$

$$f_5(x, y) = (x^0 \wedge y^1) \vee (x^1 \wedge y^1) = (\bar{x} \wedge y) \vee (x \wedge y) = (\bar{x} \vee x) \wedge y = 1 \wedge y = y \quad \text{(DCF)}$$

$$f_3(x, y) = (x^0 \vee y^0) \wedge (x^0 \vee y^1) = (x \vee y) \wedge (x \vee \bar{y}) = x \wedge (y \vee \bar{y}) = x \wedge 1 = x \quad \text{(CCF)}$$

$$\begin{aligned} f_0(x, y) &= (x^0 \vee y^0) \wedge (x^0 \vee y^1) \wedge (x^1 \vee y^0) \wedge (x^1 \vee y^1) \\ &= (x \vee y) \wedge (x \vee \bar{y}) \wedge (\bar{x} \vee y) \wedge (\bar{x} \vee \bar{y}) = ((x \vee y) \wedge (x \vee \bar{y})) \wedge ((\bar{x} \vee y) \wedge (\bar{x} \vee \bar{y})) \\ &= (x \vee (y \wedge \bar{y})) \vee (\bar{x} \vee (y \wedge \bar{y})) = (x \vee 0) \wedge (\bar{x} \vee 0) = x \wedge \bar{x} = 0 \quad \text{(CCF)} \end{aligned}$$

Remarks:

- The Boolean function f_0 doesn't have a disjunctive canonical form.
- The Boolean function $f_{2^{2^n}-1}$ doesn't have a conjunctive canonical form.

MINTERMS AND MAXTERMS

Definition 3.

Let $f: (B_2)^n \rightarrow B_2, n \in \mathbb{N}^*$ be a Boolean function of n variables.

- A conjunction of variables is called a monom.
- A monom which contains all n variables is called *canonical monom* or minterm of n variables and has the form: $x_1^{\alpha_1} \wedge \dots \wedge x_n^{\alpha_n}, \alpha_i \in B_2$.
- A disjunction containing all n variables: $x_1^{\alpha_1} \vee \dots \vee x_n^{\alpha_n}, \alpha_i \in B_2$ is called maxterm of n variables.

Example 9. For $n=3$:

- $x_1 \wedge x_2$ is a monom, but not a canonical monom;
 - $x_1 \wedge \bar{x}_2 \wedge x_3$ is a canonical monom (minterm) of 3 variables;
 - $\bar{x}_1 \vee \bar{x}_2 \vee x_3$ is a maxterm of 3 variables.
- 

MINTERMS AND MAXTERMS - PROPERTIES

Properties:

- $\forall n \in \mathbb{N}^+$, there are 2^n maxterms, denoted by $M_0, M_1, \dots, M_{2^n-1}$ and

2^n minterms, denoted by $m_0, m_1, \dots, m_{2^n-1}$.

- A maxterm is a Boolean function which is 0 for only one argument.
- A minterm is a Boolean function which is 1 for only one argument.

Remark:

- The index of a minterm/*maxterm* of n variables is obtained by the conversion in decimal of the binary number composed of the digits/*dual of the digits*, digits representing the powers of all n variables from the expression of the minterm/*maxterm*.

Example 5.

There are $2^{2^2} = 2^4 = 16$ Boolean functions of 2 variables

x	y	$f_0(x,y)$	$f_1(x,y)$	$f_2(x,y)$	$f_3(x,y)$	$f_4(x,y)$	$f_5(x,y)$	$f_6(x,y)$	$f_7(x,y)$
		<i>contradiction</i>	$x \wedge y$	$x \wedge \bar{y}$	x	$\bar{x} \wedge y$	y	$x \oplus y$	$x \vee y$
0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	1	1	1	1
1	0	0	0	1	1	0	0	1	1
1	1	0	1	0	1	0	1	0	1

x	y	$f_8(x,y)$	$f_9(x,y)$	$f_{10}(x,y)$	$f_{11}(x,y)$	$f_{12}(x,y)$	$f_{13}(x,y)$	$f_{14}(x,y)$	$f_{15}(x,y)$
		$x \downarrow y = \bar{x} \wedge \bar{y}$	$x \leftrightarrow y$ $x \oplus y$	$\bar{y}$	$y \rightarrow x = \bar{y} \vee x$	$\bar{x}$	$x \rightarrow y = \bar{x} \vee y$	$x \uparrow y = \bar{x} \vee \bar{y}$	<i>tautology</i>
0	0	1	1	1	1	1	1	1	1
0	1	0	0	0	0	1	1	1	1
1	0	0	0	1	1	0	0	1	1
1	1	0	1	0	1	0	1	0	1

EXAMPLE 10

For $n=2$, there are $2^n = 2^2 = 4$ minterms and maxterms.

x	y	$m_0 = \bar{x} \wedge \bar{y}$	$m_1 = \bar{x} \wedge y$	$m_2 = x \wedge \bar{y}$	$m_3 = x \wedge y$	$M_0 = x \vee y$	$M_1 = x \vee \bar{y}$	$M_2 = \bar{x} \vee y$	$M_3 = \bar{x} \vee \bar{y}$
0	0	1	0	0	0	0	1	1	1
0	1	0	1	0	0	1	0	1	1
1	0	0	0	1	0	1	1	0	1
1	1	0	0	0	1	1	1	1	0

We remark that the minterms have a unique value 1 on their columns,
and the maxterms have a unique value 0 on the corresponding columns.

$$m_0 = m_{00(2)} = x^0 \wedge y^0 = \bar{x} \wedge \bar{y}$$

$$m_3 = m_{11(2)} = x^1 \wedge y^1 = x \wedge y$$

$$M_1 = M_{01(2)} = x^{\bar{0}} \vee y^{\bar{1}} = x \vee \bar{y}$$

$$M_2 = M_{10(2)} = x^{\bar{1}} \vee y^{\bar{0}} = \bar{x} \vee y$$



EXAMPLE 11

For $n=3$, there are $2^n = 2^3 = 8$ minterms, denoted by $m_0, m_1, \dots, m_7$.

and 8 maxterms denoted by $M_0, M_1, \dots, M_7$

$$m_0 = m_{000(2)} = x^0 \wedge y^0 \wedge z^0 = \bar{x} \wedge \bar{y} \wedge \bar{z}$$

$$M_0 = M_{000(2)} = x^{\bar{0}} \vee y^{\bar{0}} \vee z^{\bar{0}} = x \vee y \vee z$$

$$m_3 = m_{011(2)} = x^0 \wedge y^1 \wedge z^1 = \bar{x} \wedge y \wedge z$$

$$M_3 = M_{011(2)} = x^{\bar{0}} \vee y^{\bar{1}} \vee z^{\bar{1}} = x \vee \bar{y} \vee \bar{z}$$

$$m_7 = m_{111(2)} = x^1 \wedge y^1 \wedge z^1 = x \wedge y \wedge z$$

$$M_7 = M_{111(2)} = x^{\bar{1}} \vee y^{\bar{1}} \vee z^{\bar{1}} = \bar{x} \vee \bar{y} \vee \bar{z}$$



Theorem 5.

- The conjunction of two distinct minterms is 0:

$$m_i \wedge m_j = 0, \quad \forall i \neq j, \quad i, j = 0, \dots, 2^{n-1}.$$

- The disjunction of two distinct maxterms is 1:

$$M_i \vee M_j = 1, \quad \forall i \neq j, \quad i, j = 0, \dots, 2^{n-1}.$$

- A minterm and maxterm with the same index are inverse functions.

$$M_i = \overline{m_i}, \quad \overline{M_i} = m_i, \quad \forall i = 0, \dots, 2^{n-1}.$$

Remarks:

- The conjunctive canonical form, CCF, is the conjunction of the maxterms corresponding to the values 0 of the function.
- The disjunctive canonical form, DCF, is the disjunction of the minterms corresponding to the values 1 of the function.



EXAMPLE 12

WRITE THE CANONICAL FORMS OF THE FUNCTION.

x_1	x_2	x_3	$f(x_1, x_2, x_3)$	M_2	M_4	M_5	m_0	m_1	m_3	m_6	m_7
0	0	0	1	1	1	1	1	0	0	0	0
0	0	1	1	1	1	1	0	1	0	0	0
0	1	0	0	0	1	1	0	0	0	0	0
0	1	1	1	1	1	1	0	0	1	0	0
1	0	0	0	1	0	1	0	0	0	0	0
1	0	1	0	1	1	0	0	0	0	0	0
1	1	0	1	1	1	1	0	0	0	1	0
1	1	1	1	1	1	1	0	0	0	0	1

The CCF is built by using the maxterms corresponding to the values 0 of the function.

$$CCF(f) = M_2 \wedge M_4 \wedge M_5 = (x_1 \vee \bar{x}_2 \vee x_3) \wedge (\bar{x}_1 \vee x_2 \vee x_3) \wedge (\bar{x}_1 \vee x_2 \vee \bar{x}_3)$$

The DCF is built by using the minterms corresponding to the values 1 of the function.

$$DCF(f) = m_0 \vee m_1 \vee m_3 \vee m_6 \vee m_7 =$$

$$= (\bar{x}_1 \wedge \bar{x}_2 \wedge \bar{x}_3) \vee (\bar{x}_1 \wedge \bar{x}_2 \wedge x_3) \vee (\bar{x}_1 \wedge x_2 \wedge x_3) \vee (x_1 \wedge x_2 \wedge \bar{x}_3) \vee (x_1 \wedge x_2 \wedge x_3)$$



matematician, extraordinar om de știință și
profesor, înzestrat cu un deosebit simț al umorului

https://www.youtube.com/watch?v=vALliqzw_SU

Eu sunt omul care demonstrez, nu conving.

- Ține primul curs de algebră modernă din România, *Logica și teoria demonstrației*
- A extins în spațiul cu mai multe dimensiuni derivata areolară a lui [Pompeiu](#) și a studiat funcțiile monogene de o variabilă hipercomplexă, cu aplicații la mecanică.
- A introdus algebre numite de el *Lukasiewicz trivalente și polivalente* (numite astăzi **algebre Lukasiewicz-Moisil**) și le-a întrebuințat în logică și în studiul circuitelor de comutație.
- A elaborat metode noi de analiză și sinteză a automatelor finite și a avut contribuții valoroase în domeniul teoriei algebrice a mecanismelor automate.
- A sprijinit realizarea primelor [calculatoare](#) românești, contribuind la instalarea [primului calculator de construcție românească](#) în 1957 la [Institutul de Fizică Atomică](#)
- În 1996, societatea internațională a inginerilor în electronica [IEEE](#) i-a acordat distincția rară de *Computer Pioneer Award* (post-mortem)

MOISIL – CITATE CELEBRE

- Logica ne dă mijloace de a insufla altora convingeri pe cari nu le avem.
- Omul trebuie să învețe logica nu pentru a învăța să judece corect, ci pentru a învăța mai repede să judece corect.
- Nu vom încerca să fim consecvenți; ne vom mărgini să fim logici.
- Pentru a putea întrebuința calculatorul la studiul problemelor concrete, omul e obligat să învețe să gândească exact și abstract.
- Tot ce e gândire corectă este sau matematică sau susceptibilă de matematizare.
- Matematica o văd ca ceva larg care se întinde de la filozofie la inginerie.
- Pe măsură ce se maturizează, o știință se matematizează.
- Știința e formată numai din afirmații și negații, dar trăirea unei științe e formată din întrebări și răspunsuri, din bănuieli și îndoieli.
- O teoremă e o scrisoare de dragoste către un necunoscut, către acela care îi prinde nu numai înțelesul, ci și toate subînțelesurile.
- Există și matematici pure, și matematici aplicate. În matematicile pure, cercetătorul face "cum trebuie, ce poate"; în matematicile aplicate, face "ce trebuie, cum poate".

MOISIL – CITATE CELEBRE

“Imaginația e și ea o sursă de informare.”

“Obiectul creează necesitatea, nu necesitatea creează obiectul.”

“Nu e de părerea ta cel ce te apreciază, ci cel ce te imită.”

“Dușmanii se recrutează dintre prieteni.”

„Numai prostia poate să n-aibă intermitențe.”

„Nimic nu costă mai mult decât neștiința.”

„Legile țării nu interzic nimănui să fie imbecil.”

„Ce este un pesimist? Un optimist bine informat.”

“Orice om are dreptul să bea un pahar de vin. Eu sunt om, deci eu am dreptul să beau un pahar de vin. Dar, bănd un pahar de vin, eu devin alt om...”

“Greu nu e să ai dreptate, greu e să convingi pe alții. Nu întotdeauna e greu, ci numai când ai dreptate. Și mai ales e greu să convingi pe cei care spun că s-au convins.”

